

Inspect Before Modeling

Project: Responsible Plant Classifier

1. Your mission

Your learning target

I can inspect row meaning missing values and repeated records.

I can document dataset source schema and limitations.

I can separate observed quality flags from justified corrections.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Python and Table Readiness. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Features, Labels, Targets. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the evidence shows was wrong.

Dataset shape, types, ranges, duplicates, and missingness determine whether modeling results can be trusted.



LEARNING TARGET A table contains 8 cm, -2 cm, missing and two copies of the same leaf photo. Propose a separate inspection for each issue.

Builder name and experiment date:

Readiness for the connected explanation

Why is elapsed trip duration in minutes a regression target even when recorded as whole numbers?

Readiness for Units conversions and data provenance

How many minutes are 180 seconds?

My readiness explanation

This week's end goal

By the end, I can explain inspect before modeling using a traced example and a stated limitation.

2. Notice the evidence

PICTURE ENTRY

Notice → Predict → Explain

Begin with visible evidence; do not explain more than the picture and information support.



MYSTERY The team must use inspect before modeling to solve a model-building problem. What information do they have? What important evidence is still missing?

Name three visible details without interpreting them.

Predict the data, model, or evaluation decision being made.

Identify one hidden assumption that could change the result.

State what evidence would confirm or reject your prediction.

My prediction and supporting evidence:

A closer explanation

A dataset can look like a neat table and still contain missing outcomes, repeated records or impossible values. Before fitting a model, inspect what one row represents and document where the table came from. A dataset card is a short record of its purpose, contents, collection, changes and limits. It helps another person decide whether the data suits a proposed use.

Our source is a teacher-authored fictional export called Robot practice v1, created for this lesson. Each row is intended to describe one trip, with one unique trip ID and elapsed duration in minutes. Durations may be zero or positive; None means unknown. All trips are described as indoor practice trips. No real robot measurements or personal records were collected, and there is no evidence about outdoor performance.

Trip ID · Duration in minutes · What is visible

Trip ID: A

Duration in minutes: 8

What is visible: A recorded duration

Trip ID: B

Duration in minutes: None

What is visible: An unknown duration

Trip ID: B

Duration in minutes: None

What is visible: The same ID appears again

Trip ID: C

Duration in minutes: -2

What is visible: A negative duration

Trip ID: D

Duration in minutes: 12

What is visible: A recorded duration

Keep this raw version unchanged. At this stage a flag means investigate, not automatically delete or replace. A repeated ID might be a copied row or a problem with the supposed one-row-per-trip definition. We need evidence before deciding what happened.

Apply the connected idea: Units conversions and data provenance

Two fictional activity logs use different duration units. Log A records 180 seconds and Log B records 4 minutes. Keep the raw values, units and source labels; derive a shared unit so another learner can reproduce the comparison.

3. See how it works

Input → Model → Evidence

Trace the information, fitted behavior, and evaluation proof separately.

IN

INPUT

Produce a data-quality report before fitting any model.

ML

MODEL

Dataset shape, types, ranges, duplicates, and missingness determine whether modeling results can be trusted.



OUT

EVIDENCE

Produce a data-quality report before fitting any model.

QUESTION Produce a data-quality report before fitting any model.

MODEL IDEA Dataset shape, types, ranges, duplicates, and missingness determine whether modeling results can be trusted.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

UNDER THE HOOD Dataset shape is 40 rows × 5 columns. Inspection finds 3 duplicate rows and 4 missing light values.

Draw a fresh data → model → prediction → evaluation trace:

A closer explanation

```
rows = [("A", 8), ("B", None), ("B", None),
        ("C", -2), ("D", 12)]

ids = set()
missing = 0
negative = 0

for trip_id, minutes in rows:
    ids.add(trip_id)

    if minutes is None:
        missing = missing + 1

    elif minutes < 0:
        negative = negative + 1
```

```
print(len(rows), len(ids), missing, negative)
```

A set stores each distinct ID once. The output is 5 4 2 1: five rows, four distinct IDs, two rows with unknown duration and one row with a negative duration. There is one extra occurrence beyond the four distinct IDs. This is not a count of all rows in repeated-ID groups: both B rows belong to that group.

Turn an observation into a documented action

Finding · What it establishes · Next action

Finding: B appears twice

What it establishes: Identity needs review

Next action: Check export history

Finding: Two None values

What it establishes: 2 of 5 rows lack duration

Next action: Keep unknown explicit

Finding: C has minus 2

What it establishes: Invalid elapsed duration

Next action: Check source and units

Flags can overlap: both rows with ID B also have missing duration. Do not add flag counts as if they were separate groups.

The code assumes every row is a pair and each duration is a number or None. It is a small audit, not full schema validation. It does not check text values, units, source truth or whether IDs refer to real unique events. A successful run does not certify data quality.

Apply the connected idea: Units conversions and data provenance

```
rows = [{"source": "A", "value": 180, "unit": "s"},
        {"source": "B", "value": 4, "unit": "min"}]
minutes = []
for row in rows:
    if row["unit"] == "s":
        minutes.append(row["value"] / 60)
    elif row["unit"] == "min":
```

```
minutes.append(row["value"])

else:

    minutes.append(None)

print(minutes)
```

Follow the evidence

Step or case · Evidence · Meaning

Step or case: A raw

Evidence: 180 seconds

Meaning: 3.0 minutes

Step or case: B raw

Evidence: 4 minutes

Meaning: 4 minutes

Read each row from left to right and connect the evidence to the stated rule.

The script prints [3.0, 4]. The slash operator returns a floating-point result. The original dictionaries are unchanged. Assume nonnegative numeric raw values have already been validated. An unsupported unit produces None here and must be reported for review rather than included in arithmetic.

Convert before comparing

A minute contains 60 seconds. To express 180 seconds in minutes, divide 180 by 60 to get 3 minutes. The other value is already 4 minutes, so leave it at 4. Now both quantities use the same unit and can be compared. Keep the original number, unit and source beside the converted number. That record lets another person check whether you converted the correct measurement in the correct direction.

Connect the picture and the explanation

Point to the input, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

4. Compare the cases

EVALUATION DESIGN

Four Tests, Four Kinds of Evidence

A system is not understood until it survives more than one comfortable example.



NORMAL CASE

Expected data from the intended setting



BOUNDARY CASE

Value near a threshold or class boundary



MISSING CASE

Unknown field, label, or necessary evidence



SHIFTED CASE

New subgroup, device, condition, or time

NORMAL Expected data and expected behavior.

BOUNDARY Value close to a threshold, ambiguous label, or uncertain prediction.

MISSING Necessary field, label, source, or measurement is unavailable.

SHIFTED New lighting, species, device, subgroup, or time condition.

RELEASE RULE Passing one case never proves generalization. Predefine expected behavior for all four categories.

Which case most threatens today's claim, and why?

Change one thing in the pictured case

Change: The teacher confirms B's second row is an accidental duplicate.

Prediction: Remove that duplicate with a recorded reason; C still needs correction or explicit exclusion.

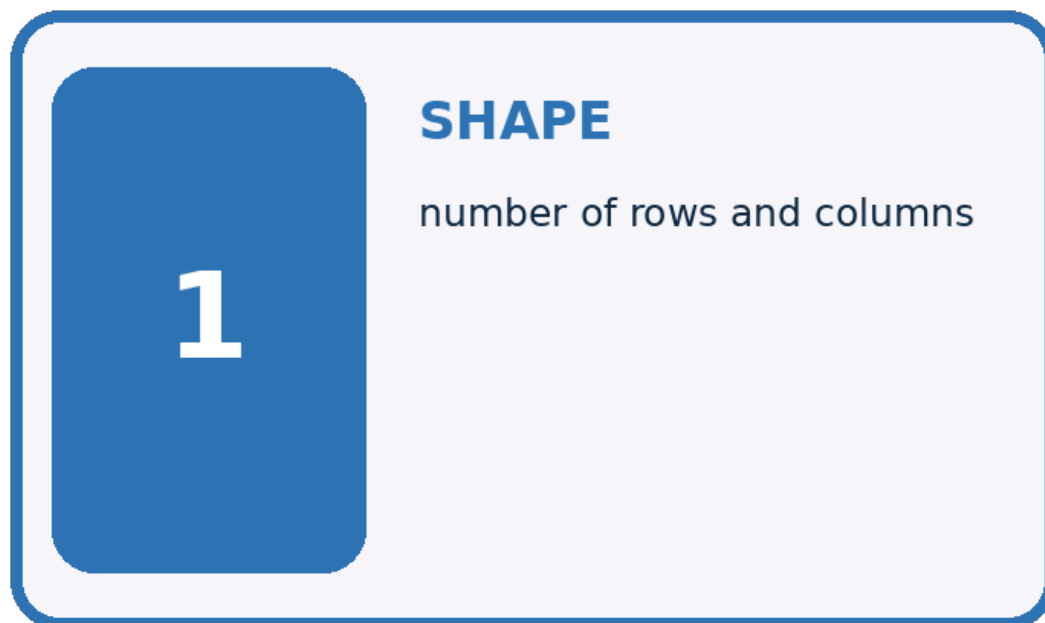
Reason: Resolving one defect does not resolve a separate impossible measurement.

5. Words that unlock the idea

VOCABULARY

Words You Can Calculate and Defend

A definition matters when it lets you interpret a new dataset, model, or result.



2

RANGE

smallest to largest plausible value

3

DUPLICATE

repeated example or record

4

MISSINGNESS

where values are absent

Term · Precise meaning · My ML example

Term: shape

Precise meaning: number of rows and columns

My ML example: _____

Term: range

Precise meaning: smallest to largest plausible value

My ML example: _____

Term: duplicate

Precise meaning: repeated example or record

My ML example: _____

Term: missingness

Precise meaning: where values are absent

My ML example: _____

MISCONCEPTION TO REPAIR A model will automatically ignore harmless data-quality problems.

Repair the claim using a definition, calculation, or test record:

Words used in the closer explanation

Word · Meaning

Word: Provenance

Meaning: Where data came from and how it was created

Word: Schema

Meaning: The stated meaning types units and rules for fields

Word: Missing value

Meaning: An unknown or unavailable value rather than a measured zero

Word: Dataset card

Meaning: A concise record of dataset contents creation intended use and limitations

Terms for Units conversions and data provenance

Term · Meaning

Term: Provenance

Meaning: The origin and processing history of data

Term: Conversion

Meaning: Expressing a quantity in another unit

Term: Raw value

Meaning: The value as originally recorded

Term: Derived value

Meaning: A value calculated from recorded data

6. Follow a worked example

WORKED EVIDENCE

Reason Step by Step

Predict first. Then check each step. A correct answer without visible evidence cannot be audited.

WEEK 5 • WORKED EVIDENCE

- 1 Dataset shape is 40 rows × 5 columns.
- 2 Inspection finds 3 duplicate rows and 4 missing light values.
- 3 Leaf widths range from 0.4 to 48 cm; inspect the extreme value and unit.
- 4 Document, repair, or exclude by a stated rule—never silently delete inconvenient rows.

STEP 1 Dataset shape is 40 rows × 5 columns.

STEP 2 Inspection finds 3 duplicate rows and 4 missing light values.

STEP 3 Leaf widths range from 0.4 to 48 cm; inspect the extreme value and unit.

STEP 4 Document, repair, or exclude by a stated rule—never silently delete inconvenient rows.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a score.

Recalculate, retrace, or restate the evidence independently:

A closer explanation

For Robot practice v1, record the purpose as practising inspection, the source as teacher-authored fictional data, and the row meaning as an intended single indoor trip. Record the five-row count, the ID field,

minutes as the duration unit and None as unknown. List the duplicate-ID and invalid-duration findings. State that no cleaning has occurred, no train/test split is assigned, and no model has been evaluated.

Add intended use, excluded uses, known gaps, version and a contact or owner. Here the lesson teacher maintains the practice copy; its allowed classroom purpose is inspection practice. Do not describe it as validated navigation data. For a real external dataset, also verify its actual permission or licence and collection context; an absent field in the documentation is an unanswered question, not permission to invent a value.

What fraction of the five raw rows has unknown duration? Can you subtract two missing rows, one negative row and one extra ID occurrence from five to count distinct usable trips? Explain using the B rows.

Apply the connected idea: Units conversions and data provenance

Convert 90 seconds and 0 seconds. Which is longer: 90 seconds or 2 minutes?

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Inspect Before Modeling

Spot a duplicate, an impossible value and a valid boundary before fitting.

The facts we will use

Toy heights table: A=3, B=5, B=5, C=-2, D=0 cm. Each plant should have exactly one nonnegative height record.

All inputs and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-4/5

Inspect Before Modeling

Start with the supplied facts
Follow each arrow. Ask which fact allows the next step.



1. Count rows

Five raw rows represent four distinct IDs; B is duplicated.





2. Check schema

C violates the nonnegative-height rule; D=0 is valid under this contract.





3. Flag before fitting

Flag duplicate B and invalid C. Do not count B twice or silently replace C.

CHANGE ONE THING
The teacher confirms B's second row is an accidental duplicate.

Explain the changed result using the matching lesson.

Read the picture one step at a time

Step · What happens and why

Step: 1. Count rows

What happens and why: Five raw rows represent four distinct IDs; B is duplicated.

Step: 2. Check schema

What happens and why: C violates the nonnegative-height rule; D=0 is valid under this contract.

Step: 3. Flag before fitting

What happens and why: Flag duplicate B and invalid C. Do not count B twice or silently replace C.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Read the picture in order. Flag duplicate B and invalid C. Do not count B twice or silently replace C. The arrows show which recorded input or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

What this example does not establish

A plausible nonnegative value can still be incorrectly measured; schema checks do not establish truth.

Read the labelled picture: Which original information should the inspection log preserve? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

Change one thing: The teacher confirms B’s second row is an accidental duplicate. Predict the result before checking the explanation. What remains the same?

7. Find and repair a mistake

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, data problem, or experiment error.



COMPARE • DIAGNOSE • REPAIR

Tempting claim: A model will automatically ignore harmless data-quality problems. Inspect the dataset, split, calculation, error type, subgroup, or experiment record; identify one testable cause; change one factor; rerun the same evidence.

Evidence record · Expected · Actual · Likely cause / next test

Evidence record: normal case

Expected: passes

Actual: _____

Likely cause / next test: _____

Evidence record: boundary case

Expected: defined behavior

Actual: _____

Likely cause / next test: _____

Evidence record: missing case

Expected: safe fallback

Actual: _____

Likely cause / next test: _____

Evidence record: shifted case

Expected: measured limit

Actual: _____

Likely cause / next test: _____

CLAIM TO CHALLENGE A model will automatically ignore harmless data-quality problems.

Diagnose one cause and name the cleanest next experiment:

A closer explanation

A complete-looking card does not prove the table is accurate or suitable for a model. It makes claims and unanswered questions reviewable. Report raw counts separately from any later accepted subset, with explicit rules for missing, invalid and repeated records. Do not silently change denominators after cleaning.

For this exercise, zero is allowed and unknown is not zero. That is a stated schema choice, not a universal rule for all duration datasets. In a real task, measurement resolution and event definitions determine which values make sense. Dataset inspection also does not replace protecting held out data from model choices.

A card says the dataset is ready because it has no empty cells. Name two checks that claim

misses, then write a more accurate statement about Robot practice v1.

Apply the connected idea: Units conversions and data provenance

Correct conversion does not establish that the sources measured the same activity, used the same start and stop rules, or are accurate. Provenance helps someone investigate those differences.

A record contains value 8 and unit ticks with no conversion rule. Should you assume seconds?

8. Practise together

ML STUDIO

Specify Before You Run

A trustworthy result can be reproduced from the recorded data, code, settings, and evaluation rule.



FREEZE THE EXPERIMENT

Record question, intended use, data version, grouped split, preprocessing, features, model settings, seed, metric, predictions, errors, subgroup results, and human review before claiming improvement.

Experiment record

QUESTION + intended use + success threshold

DATA version + grouped split + leakage check

CODE/preprocessing + features + model + settings + seed

METRICS + positive class + units + denominators

PREDICTIONS + errors + subgroup slices + limitations

HUMAN reviewer + release, revise, limit, or stop decision

Write the minimum record needed to reproduce today's result:

A closer explanation

A learner replaces both None values with 0 and removes the negative sign from -2. The table now passes a simple nonnegative check, but those edits invent durations. Keep the original values, record the reason for each flag, and check the source. If a correction is supported, save a new version with the original value, corrected value and reason. If not, keep the uncertainty visible.

Another learner says all robot trips are covered because every row says indoor. A dataset card should instead state that outdoor trips are absent and that the data is fictional. Extra copies of existing rows do not create new conditions or new independent evidence.

Apply the connected idea: Units conversions and data provenance

Comparing raw 180 to raw 4 ignores their units. Dividing 4 by 60 without checking its unit is also wrong: it was already minutes. Branch on the documented unit and keep a conversion note.

Practise with less help: Interpret the audit counts

Use workbook W2 for the connected example “Dataset inspection and cards”. First fill the input and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan · Do · Check

Plan: What is given? Which rule or procedure applies?

Do: Record each intermediate step.

Check: Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Dataset inspection and cards. Proposed explanation: “A complete card certifies data quality”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Practise the connected topic: Units conversions and data provenance

Use its named workbook W2. Identify the given inputs, try the middle steps with less help, then explain your trace. Check this claim before group discussion: A successful run proves the conclusion

Explain the picture

Which original information should the inspection log preserve? Point to the relevant step in the picture, then write the changed result and the rule or evidence that supports it.

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

9. Run the investigation

EVIDENCE LAB

Predict → Run → Calculate → Interpret

Keep the test conditions fixed, record every result, and change only one planned factor.

LAB

PREDICT → RUN → CALCULATE → INTERPRET

Data Triage: calculate missing counts, flag ranges, find duplicates, and write a keep/repair/exclude reason for each issue.

1. Write the hypothesis and expected evidence before running.
2. Data Triage: calculate missing counts, flag ranges, find duplicates, and write a keep/repair/exclude reason for each issue.
3. Run one normal, one boundary, one missing, and one shifted case.
4. Calculate the metric from raw counts or row-level errors.
5. Interpret what changed, what did not, and what remains unsafe or unknown.

Case · Prediction · Actual · Calculation / evidence · Meaning / next test

Case: Normal

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Boundary

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Missing

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Case: Shifted

Prediction: _____

Actual: _____

Calculation / evidence: _____

Meaning / next test: _____

Plan → try → check

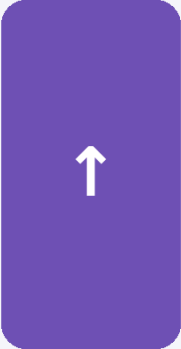
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

10. Build your understanding

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce evidence without copying the example.



ML BUILDER MASTERY
1 Define • 2 Calculate • 3 Trace code • 4 Run experiment • 5 Diagnose errors • 6 Improve one cause • 7 Defend limits

DEFINE Use all four terms precisely: shape, range, duplicate, missingness.

CALCULATE Reproduce the worked evidence with units and denominator.

TRACE Explain the data, code or decision path.

DIAGNOSE A model will automatically ignore harmless data-quality problems.

TRANSFER Solve a fresh dataset or model case.

IMPROVE Change one cause and rerun fixed evidence.

DEFEND State intended use, metric, limitation, and human responsibility.

MASTERY RULE A coached calculation becomes mastery only after a different case is completed independently.

11. Connect your project

CUMULATIVE PROJECT

Responsible Plant Classifier

Every week adds one reviewable artifact to the same evidence-based capstone.

**RESPONSIBLE PLANT CLASSIFIER**

Audit the Plant Classifier table and attach a data-quality report before any baseline or model is trained.

THIS WEEK'S CHECKPOINT Audit the Plant Classifier table and attach a data-quality report before any baseline or model is trained.

Which requirement or risk does this artifact address?

Which dataset, split, code, and metric version does it use?

What normal and shifted cases were tested?

What remains uncertain or weak?

What decision still belongs to a teacher or qualified human?

RESPONSIBLE RELEASE Never use this classroom prototype for medical, pesticide, agricultural, or other consequential decisions.

My evidence-linked project decision:

Evidence for your project

For your Responsible Plant Classifier, save the input, procedure, observed result and limitation of this check. Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

12. Show what you know

INDEPENDENT PROOF

Explain • Calculate • Transfer • Defend

Complete this with a fresh case after guided practice and compare it with the mastery criteria.



INDEPENDENT ML BUILDER PROOF
Produce a data-quality report before fitting any model.

REQUIRED PROOF Produce a data-quality report before fitting any model.

- ☐ Define the task, feature/target roles, and intended-use boundary.
- ☐ Show the calculation, code trace, or experiment record.
- ☐ Test a normal case and one revealing boundary, missing, or shifted case.
- ☐ Diagnose one failure without changing several factors at once.
- ☐ Name a limitation, fairness concern, and the human-review action.

Independent evidence and defense:

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach and defend it

NEXT BRIDGE Carry the experiment evidence into: Label Quality.

13. Study a complete case

Week 5 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

Lengths include 5 cm, blank and -3 cm, plus a duplicate photo row.

The question

Name checks and a deletion risk.

Worked explanation

Check missingness, physical validity, units and duplicate identity. Investigate and document repair. Blind deletion can disproportionately remove a relevant condition.

Explain it in your own words

Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Trace the supplied case visually

Inspect data before choosing repairs

Evidence or stage · Value or result

Evidence or stage: 5 cm

Value or result: Physically plausible

Evidence or stage: Blank

Value or result: Missing measurement

Evidence or stage: -3 cm

Value or result: Invalid length

Evidence or stage: Duplicate photo

Value or result: Check record identity

Lengths include 5 cm, blank and -3 cm, plus a duplicate photo row.

Name checks and a deletion risk.

Check missingness, physical validity, units and duplicate identity. Investigate and document repair. Blind deletion can disproportionately remove a relevant condition.

14. Try a new case independently

Independent case: Inspect and document independently

A new fictional indoor trip export has (P,4), (Q,None), (Q,None), (R,-1), (S,6), (T,0), with durations in minutes and the same schema. Count raw rows, distinct IDs, missing durations, negative durations and extra ID occurrences. Identify overlap. Draft a short dataset card stating source, row meaning, unit, findings, unchanged version and one excluded use.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Week 5 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A table has height_cm values 12, 15, -8 and a blank. Identify two issues and a suitable first action.

My answer, with a trace, calculation or supporting evidence

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed input and expected result

Write or draw your reasoning and evidence.

Connect to your project

Audit the Plant Classifier table and attach a data-quality report before any baseline or model is trained.

The evidence I will keep

Write or draw your reasoning and evidence.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice evidence together.

Transfer practice from the original programme

A table contains 8 cm, -2 cm, missing and two copies of the same leaf photo. Propose a separate inspection for each issue.

Explain the mechanism, evidence and one limit.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed. A later check uses a changed case, not a copied answer.

15. Your lesson route

Start with this question: Spot a duplicate, an impossible value and a valid boundary before fitting.

By the end, I can · How I will show it

By the end, I can: Inspect row meaning missing values and repeated records
How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Document dataset source schema and limitations
How I will show it: Explain the deciding step, show a trace or test, and state one limit.

By the end, I can: Separate observed quality flags from justified corrections

How I will show it: Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project evidence. Your teacher may spread this lesson across several classes.

Keep your first prediction. Compare it with what the evidence shows and explain any change.

16. Finish and return

Close your notes. Explain this goal in your own words: Inspect row meaning missing values and repeated records

My deciding step, evidence and one remaining question

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an input or assumption and justify your prediction.

Save your workbook evidence. At the next review, try a different case before opening your old answer.